MATH 3A

Spring 2026

Practice Final Solutions

1. **True or False? Give convincing reasons for your answers.**

   (a) If A is a 4×4 matrix of rank 3, then $\det(A) = 0$.

$$\boxed{\text{True}}$$

$$\operatorname{rank}(A) = 3 < 4$$

The columns of A are linearly dependent.

A is singular.

$$\boxed{\det(A) = 0}$$

   (b) If the columns of a 5×5 matrix are linearly independent, then the rows must be linearly independent too.

$$\boxed{\text{True}}$$

For a square matrix A,

$$\text{columns linearly independent} \iff \operatorname{rank}(A) = 5 \iff \text{rows linearly independent.}$$

   (c) An $n \times n$ matrix is invertible if and only if the equation $AX = 0$ has only the solution $X = 0$.

$$\boxed{\text{True}}$$

$$A \text{ invertible} \iff \operatorname{Null}(A) = \{0\} \iff AX = 0 \text{ has only the trivial solution.}$$

   (d) Let A be an arbitrary 2×3 matrix. Then the third column of A is a linear combination of the first and second columns.

$$\boxed{\text{False}}$$

A counterexample is

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The first two columns are both zero:

$$A_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

So every linear combination of the first two columns is

$$c_1A_1 + c_2A_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

But

$$A_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Therefore

$$A_3 \notin \text{Span}\{A_1, A_2\}.$$

(e) Let A be an $m \times n$ matrix. Then $\text{rank}(A) = \text{rank}(A^T)$.

$$\boxed{\text{True}}$$

$$\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A)).$$

Since the rows of A are the columns of A^T,

$$\text{rank}(A^T) = \dim(\text{Row}(A)).$$

Thus

$$\boxed{\text{rank}(A) = \text{rank}(A^T)}.$$

(f) If A is $n \times m$ and $\text{Null}(A) = \{0\}$, then $n = m$ and A is invertible.

$$\boxed{\text{False}}$$

A counterexample is

$$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A : \mathbb{R} \to \mathbb{R}^2.$$

Then

$$Ax = 0 \iff \begin{bmatrix} x \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \iff x = 0.$$

So

$$\text{Null}(A) = \{0\}.$$

But

$$A \text{ is } 2 \times 1, \qquad n \neq m,$$

and A is not a square matrix, so it is not invertible.

(g) For any square matrix A, we have $\det(2A) = 2\det(A)$.

$$\boxed{\text{False}}$$

If A is $n \times n$, then

$$\det(2A) = 2^n \det(A).$$

For example, take

$$A = I_2.$$

Then

$$\det(2I_2) = \det \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 4,$$

but

$$2\det(I_2) = 2.$$

Thus

$$\det(2A) \neq 2\det(A)$$

in general.

(h) Suppose A is a 5×5 matrix with $\det(A) = 0$. Then the equation $AX = 0$ must have infinitely many solutions.

$$\boxed{\text{True}}$$

$$\det(A) = 0 \Longrightarrow A \text{ is singular} \Longrightarrow \mathrm{Null}(A) \neq \{0\}.$$

Thus there is a nonzero vector v such that

$$Av = 0.$$

Then for every scalar t,

$$A(tv) = tAv = 0.$$

So

$$\boxed{AX = 0 \text{ has infinitely many solutions.}}$$

(i) If $A + I$ is invertible, then -1 is not an eigenvalue of A.

$$\boxed{\text{True}}$$

If -1 were an eigenvalue of A, then for some nonzero vector v,

$$Av = -v.$$

Then

$$(A + I)v = Av + v = -v + v = 0.$$

So $A + I$ would have a nonzero vector in its null space, hence would not be invertible. Therefore

$$\boxed{-1 \text{ is not an eigenvalue of } A.}$$

(j) If $\det(A - A^2) = 0$, then either 0 or 1 is an eigenvalue of A.

$$\boxed{\text{True}}$$

$$A - A^2 = A(I - A).$$

Therefore

$$0 = \det(A - A^2) = \det(A)\det(I - A).$$

Hence

$$\det(A) = 0 \qquad \text{or} \qquad \det(I - A) = 0.$$

If

$$\det(A) = 0,$$

then 0 is an eigenvalue of A.
If

$$\det(I - A) = 0,$$

then

$$\det(A - I) = 0,$$

so 1 is an eigenvalue of A.

(k) A matrix is invertible if and only if 0 is not one of its eigenvalues.

$$\boxed{\text{True}}$$

$$0 \text{ is an eigenvalue of } A \iff Av = 0 \text{ for some } v \neq 0.$$

This is equivalent to

$$\mathrm{Null}(A) \neq \{0\}.$$

Thus

$$0 \text{ is not an eigenvalue} \iff \mathrm{Null}(A) = \{0\} \iff A \text{ is invertible.}$$

(l) Similar matrices must have the same determinant.

$$\boxed{\text{True}}$$

If A and B are similar, then

$$B = PAP^{-1}$$

for some invertible matrix P. Hence

$$\det(B) = \det(PAP^{-1}) = \det(P)\det(A)\det(P^{-1}).$$

Since

$$\det(P^{-1}) = \frac{1}{\det(P)},$$

we get

$$\boxed{\det(B) = \det(A)}.$$

(m) Similar matrices must have the same rank.

$$\boxed{\text{True}}$$

If

$$B = PAP^{-1},$$

then multiplication by the invertible matrices P and P^{-1} does not change rank. Hence

$$\boxed{\operatorname{rank}(B) = \operatorname{rank}(A)}.$$

(n) The matrices

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 3 \\ -1 & 0 \end{bmatrix}$$

are similar.

$$\boxed{\text{True}}$$

For A,

$$\operatorname{tr}(A) = 4, \qquad \det(A) = 3.$$

Thus

$$p_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

For B,

$$\operatorname{tr}(B) = 4, \qquad \det(B) = 3.$$

Thus

$$p_B(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

Both matrices have two distinct eigenvalues:

$$\lambda = 1, 3.$$

Therefore both matrices are diagonalizable and both are similar to

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

Hence

$$\boxed{A \text{ and } B \text{ are similar.}}$$

(o) The matrices

$$A = \begin{bmatrix} -3 & 1 \\ 3 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$$

are similar.

$$\boxed{\text{False}}$$

Similar matrices must have the same trace. But

$$\operatorname{tr}(A) = -3 + 2 = -1,$$

while

$$\operatorname{tr}(B) = 1 + 0 = 1.$$

Since

$$\operatorname{tr}(A) \neq \operatorname{tr}(B),$$

we get

$$\boxed{A \text{ and } B \text{ are not similar.}}$$

(p) A vector X is called an eigenvector of a matrix A if there is a number λ such that $AX = \lambda X$.

$$\boxed{\text{False}}$$

The missing condition is

$$X \neq 0.$$

For example, the zero vector satisfies

$$A0 = \lambda 0$$

for every scalar λ, but the zero vector is not an eigenvector.
The correct definition is:

$$X \neq 0, \qquad AX = \lambda X.$$

(q) The eigenvalues and eigenvectors of A and A^T are always the same.

$$\boxed{\text{False}}$$

It is true that A and A^T have the same eigenvalues, but their eigenvectors do not have to be the same.
A counterexample is

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}.$$

For A and $\lambda = 1$,

$$A - I = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix},$$

so

$$x_2 = 0, \qquad E_1(A) = \text{Span}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}.$$

For A^T and $\lambda = 1$,

$$A^T - I = \begin{bmatrix}0 & 0\\1 & 1\end{bmatrix},$$

so

$$x_1 + x_2 = 0, \qquad E_1(A^T) = \text{Span}\left\{\begin{bmatrix}1\\-1\end{bmatrix}\right\}.$$

Thus

$$E_1(A) \neq E_1(A^T).$$

(r) If two matrices have the same eigenvalues then they are similar.

$$\boxed{\text{False}}$$

A counterexample is

$$A = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, \qquad B = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}.$$

Both have characteristic polynomial

$$(\lambda - 1)^2.$$

So both have the same eigenvalue:

$$\lambda = 1.$$

But

$$A = I_2$$

is similar only to itself, because

$$PI_2P^{-1} = I_2.$$

Since

$$B \neq I_2,$$

we get

$$\boxed{A \text{ and } B \text{ are not similar.}}$$

(s) If a 4×4 matrix is diagonalizable, then it must have distinct eigenvalues.

$$\boxed{\text{False}}$$

A counterexample is

$$A = I_4.$$

Then

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is already diagonal, so it is diagonalizable.

But the only eigenvalue is

$$\lambda = 1,$$

with algebraic multiplicity 4.

(t) If a 3×3 matrix is not diagonalizable, then it must have an eigenvalue of multiplicity 3.

$$\boxed{\text{False}}$$

A counterexample is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The characteristic polynomial is

$$p_A(\lambda) = (1-\lambda)^2(2-\lambda).$$

So the eigenvalues are

$$\lambda = 1 \quad \text{with algebraic multiplicity 2,}$$

$$\lambda = 2 \quad \text{with algebraic multiplicity 1.}$$

For $\lambda = 1$,

$$A - I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

so

$$x_2 = 0, \qquad x_3 = 0.$$

Thus

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\},$$

and

$$\dim(E_1) = 1 < 2.$$

Therefore A is not diagonalizable, but no eigenvalue has multiplicity 3.

2. Calculate $\det(A)$ for

$$A = \begin{bmatrix} k & 0 & 0 \\ 0 & 3 & k \\ 0 & k & 12 \end{bmatrix}.$$

Determine the values of k for which A is invertible and find the inverse matrix A^{-1}.

Solution.

$$\det(A) = k \begin{vmatrix} 3 & k \\ k & 12 \end{vmatrix} = k(36 - k^2).$$

$$A \text{ invertible} \iff k(36-k^2) \neq 0 \iff \boxed{k \neq 0, \ \pm 6}.$$

$$\boxed{A^{-1} = \begin{bmatrix} \dfrac{1}{k} & 0 & 0 \\ 0 & \dfrac{12}{36-k^2} & \dfrac{-k}{36-k^2} \\ 0 & \dfrac{-k}{36-k^2} & \dfrac{3}{36-k^2} \end{bmatrix}}$$

3. Let

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \\ 1 & -1 & 1 & -1 \end{bmatrix}.$$

   (a) Find a basis for $\mathrm{Col}(A)$ and $\mathrm{Null}(A)$.

   (b) Find $\mathrm{rank}(A)$ and $\mathrm{nullity}(A)$ and confirm the Rank–Nullity Theorem.

   (c) Is A onto as a linear transformation? If not, give a vector that is missing from the range of A.

   (d) Consider the vector

$$v = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}.$$

   Show that v is in the column space of A. Then find the coordinates of v with respect to the basis you found for $\mathrm{Col}(A)$ in part (a).

Solution.

$$\mathrm{rref}(A) = \begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$\boxed{\mathcal{B}_{\mathrm{Col}(A)} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}}$$

$$x_1 + \frac{1}{2}x_3 = 0, \qquad x_2 - \frac{1}{2}x_3 + x_4 = 0.$$

$$x_3 = s, \qquad x_4 = t.$$

$$X = s \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\boxed{\mathcal{B}_{\mathrm{Null}(A)} = \left\{ \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}}$$

$$\boxed{\operatorname{rank}(A) = 2}, \qquad \boxed{\operatorname{nullity}(A) = 2}.$$

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = 2 + 2 = 4.$$

$$A : \mathbb{R}^4 \to \mathbb{R}^3, \qquad \operatorname{rank}(A) = 2 < 3.$$

$$\boxed{A \text{ is not onto.}}$$

A vector in the range has the form

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} a+b \\ 2a \\ a-b \end{bmatrix}.$$

$$x + z = y.$$

$$\boxed{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \notin \mathrm{Col}(A)}$$

because

$$1 + 0 \neq 0.$$

For

$$v = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix},$$

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}.$$

$$2c_1 = 1 \Rightarrow c_1 = \frac{1}{2}.$$

$$c_1 + c_2 = 3 \Rightarrow c_2 = \frac{5}{2}.$$

$$\boxed{v = \frac{1}{2} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \frac{5}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}}$$

$$\boxed{[v]_{\mathcal{B}} = \begin{bmatrix} \frac{1}{2} \\ \frac{5}{2} \end{bmatrix}}$$

4. Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

and use the result to find the inverses of

$$B = \begin{bmatrix} 3 & 3 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}, \qquad C = \begin{bmatrix} -1 & 0 & 0 \\ -1 & -1 & 0 \\ -1 & -1 & -1 \end{bmatrix}.$$

Solution.

It is not hard to show that:

$$A^{-1} = \boxed{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}$$

$$B = 3A.$$

$$B^{-1} = \frac{1}{3}A^{-1} = \boxed{\begin{bmatrix} \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{3} \end{bmatrix}}$$

$$C = -A^T.$$

$$C^{-1} = -(A^T)^{-1} = -(A^{-1})^T.$$

$$C^{-1} = \boxed{\begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}}$$

5. Consider the inhomogeneous equation

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 13 \end{bmatrix}.$$

Use Cramer's Rule to show that $x_2 = 0$. What would you do if you were not supposed to use Cramer's Rule?

Solution.

$$M = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 10 \\ 8 \\ 13 \end{bmatrix}.$$

$$\det(M) = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 3 \end{vmatrix} = -1.$$

$$M_2 = \begin{bmatrix} 2 & 10 & 2 \\ 1 & 8 & 2 \\ 2 & 13 & 3 \end{bmatrix}.$$

$$\det(M_2) = \begin{vmatrix} 2 & 10 & 2 \\ 1 & 8 & 2 \\ 2 & 13 & 3 \end{vmatrix} = 0.$$

$$x_2 = \frac{\det(M_2)}{\det(M)} = \frac{0}{-1} = \boxed{0}.$$

Without Cramer's Rule:

$$\left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 1 & 1 & 2 & 8 \\ 2 & 2 & 3 & 13 \end{array}\right] \xrightarrow{\textit{Row reduction}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

$$\boxed{x_1 = 2, \qquad x_2 = 0, \qquad x_3 = 3.}$$

6. Determine which of the following subsets are linear subspaces. For those that are subspaces, find a basis.

(a)

$$S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : y = x^2 \right\} \subseteq \mathbb{R}^2.$$

(b)

$$S = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a + b = 2c \right\} \subseteq \mathbb{R}^3.$$

(c)

$$S = \operatorname{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\} \subseteq \mathbb{R}^3.$$

(d)

$$S = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : a + b = c + d, \quad a + 2b + 3c + 4d = 0 \right\} \subseteq \mathbb{R}^4.$$

Solution.

(a)

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in S, \qquad 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \notin S.$$

$$\boxed{S \text{ is not a subspace.}}$$

(b)

$$a + b = 2c \iff a = 2c - b.$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2c - b \\ b \\ c \end{bmatrix} = b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

$$\boxed{S \text{ is a subspace.}}$$

$$\boxed{\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}}$$

(c)

$$S = \text{Span}\{v_1, v_2\}.$$

$$\boxed{S \text{ is a subspace.}}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \nparallel \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}.$$

$$\boxed{\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\}}$$

(d)

$$a + b = c + d, \qquad a + 2b + 3c + 4d = 0.$$

$$a + b - c - d = 0, \qquad a + 2b + 3c + 4d = 0.$$

Note that

$$S = \text{Null}\left(\begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \right)$$

So:

$$\boxed{S \text{ is a subspace.}}$$

One can then do row reductions and find that:

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = c \begin{bmatrix} 5 \\ -4 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 6 \\ -5 \\ 0 \\ 1 \end{bmatrix}.$$

$$\boxed{S \text{ is a subspace.}}$$

$$\boxed{\mathcal{B} = \left\{ \begin{bmatrix} 5 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -5 \\ 0 \\ 1 \end{bmatrix} \right\}}$$

7. Consider the matrix

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

(a) Find all eigenvalues of A and their algebraic multiplicities.

(b) Find a basis for each eigenspace.

(c) Determine whether A is diagonalizable.

(d) If A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

(e) Compute A^{10} using the diagonalization.

Solution.

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 0 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (4-\lambda)^2(2-\lambda).$$

$$\boxed{\lambda = 4 \text{ has algebraic multiplicity } 2}$$

$$\boxed{\lambda = 2 \text{ has algebraic multiplicity } 1}$$

For $\lambda = 4$:

$$A - 4I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix}.$$

$$x_3 = 0.$$

$$\boxed{E_4 = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}}$$

For $\lambda = 2$:

$$A - 2I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$x_1 = 0, \qquad 2x_2 + x_3 = 0.$$

$$\boxed{E_2 = \text{Span}\left\{ \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \right\}}$$

$$\dim(E_4) + \dim(E_2) = 2 + 1 = 3.$$

$$\boxed{A \text{ is diagonalizable.}}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

$$\boxed{A = PDP^{-1}.}$$

$$A^{10} = PD^{10}P^{-1}.$$

$$D^{10} = \begin{bmatrix} 4^{10} & 0 & 0 \\ 0 & 4^{10} & 0 \\ 0 & 0 & 2^{10} \end{bmatrix}.$$

$$P^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}.$$

$$A^{10} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 4^{10} & 0 & 0 \\ 0 & 4^{10} & 0 \\ 0 & 0 & 2^{10} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}.$$

$$\boxed{A^{10} = \begin{bmatrix} 4^{10} & 0 & 0 \\ 0 & 4^{10} & \dfrac{4^{10} - 2^{10}}{2} \\ 0 & 0 & 2^{10} \end{bmatrix}}$$

8. Consider the matrix

$$A = \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}.$$

(a) Find the characteristic polynomial of A.

(b) Find all eigenvalues of A over $\mathbb{C}$.

(c) Find a basis for each eigenspace.

(d) Find an invertible complex matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Solution.

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 4 \\ -4 & 3-\lambda \end{vmatrix}.$$

$$p(\lambda) = (3-\lambda)^2 + 16.$$

$$\boxed{p(\lambda) = \lambda^2 - 6\lambda + 25.}$$

$$\lambda = \frac{6 \pm \sqrt{36-100}}{2} = 3 \pm 4i.$$

$$\boxed{\lambda_1 = 3+4i, \qquad \lambda_2 = 3-4i.}$$

For $\lambda = 3+4i$:

$$A - (3+4i)I = \begin{bmatrix} -4i & 4 \\ -4 & -4i \end{bmatrix}.$$

$$-ix_1 + x_2 = 0 \Rightarrow x_2 = ix_1.$$

$$\boxed{E_{3+4i} = \mathrm{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ i \end{bmatrix} \right\}}$$

For $\lambda = 3-4i$, it is automatic from a theorem that the eigenvector is the conjugate of the eigenvector we got above, so:

$$\boxed{E_{3-4i} = \mathrm{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -i \end{bmatrix} \right\}}$$

$$P = \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}, \qquad D = \begin{bmatrix} 3+4i & 0 \\ 0 & 3-4i \end{bmatrix}.$$

$$\boxed{A = PDP^{-1}.}$$

9. Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

   (a) Find the characteristic polynomial of A.
   (b) Find all eigenvalues and their algebraic multiplicities.
   (c) Find a basis for the eigenspace corresponding to each eigenvalue.
   (d) Is A diagonalizable? Justify your answer.

Solution.

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)^3.$$

$$\boxed{p(\lambda) = (2-\lambda)^3.}$$

$$\boxed{\lambda = 2 \text{ has algebraic multiplicity } 3.}$$

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$x_2 = 0, \qquad x_3 = 0.$$

$$\boxed{E_2 = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}}$$

$$\dim(E_2) = 1 < 3.$$

$$\boxed{A \text{ is not diagonalizable.}}$$